TOM SEMRAD

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SUMMARY

AI evaluation designer and AI-native builder. I test AI systems to find out if they actually work. Before I run anything, I decide what counts as a pass and build a broken version that should fail, so I know the test can tell the difference. When something doesn't hold up, that goes on the record too. I use AI coding tools (Claude Code) to build what I test, and my part is figuring out whether it's any good. Data Science certificate (2026); psychology minor that shapes how I think about people and the systems built for them.

SELECTED RESULTS (each reproducible from a committed script)

- **Designed a retrieval evaluation that an engine developer adopted in his release gate:** an application-level test (n=127 queries) that the lead engineer of a deterministic game-AI engine ran to approve his embedder release.
- **Built and proved an NPC belief system inside Unreal Engine (5.3, now 5.8):** headless runs reproduced a pre-registered prediction within about one second; 214 NPCs with zero failed calls and a worst frame of 38 ms; then found and documented the real limit (the CPU relation judge tops out at 2 to 5 new free-text claims per second), then removed it for game events: 40 new claims per second with zero late gossip rounds.
- **Evaluated three candidate NLI models against a production judge** on a 48-pair adversarial set plus MNLI, against a fitness bar set in advance: none qualified, which saved a planned engine research spike.
- **Stopped a fine-tuned model from shipping:** measured that a DPO-trained model degraded real decisions (n=318, z = -6.27) even though it scored well on the metric it was trained against, then rebuilt the metric so it tracks real usefulness.
- **Caught my own errors before they spread:** found two tests that were passing without testing anything, and a performance diagnosis that was wrong; corrected the record. Maintain a ledger of 34 claims, each tied to a script that reproduces it, negative results included.

PROJECTS

Hearsay: provenance-aware belief layer for game NPCs (private repo, shared with a game-AI engine developer)

- NPCs believe what they hear based on who said it, where the claim started, and whether the source held up when checked; copies of one source count once; contested subjects make NPCs verify instead of act.
- Every feature shipped with a pre-registered test and a control that must fail; performance work cut decision time from 1.4 s to about 35 ms and load time from 10 minutes to under a second with identical answers.

aioos-retrieval-eval (public) · **aioos-channel-routing** (public)

- Application-level evaluations: retrieval quality on game-memory queries (nDCG@3 = 0.68, n=127), and embedding routing vs keyword rules (+0.35, n=116), each with adjudicated labels and traps built to catch shortcuts.

Orlog: AI decision tool (retired)

- Shipped a live AI product with subscription billing (Stripe) on Modal and Cloudflare. Then tested it honestly: its output tied a well-written single prompt on decision quality, so I retired it and kept the lesson.

Mimir: fine-tuned small language models · huggingface.co/donmerlin

- Built a preference-optimization (DPO) pipeline end to end: training-pair generation, GPU training on Modal, evaluation, and the measurement that kept a degrading model from deployment.

SKILLS

Evaluation: pre-registration · positive and negative controls · held-out and adversarial test sets · bootstrap confidence intervals · balanced accuracy / nDCG · catching data leaks and hollow tests **Building:** AI-assisted development (Claude Code) · Python (read, test, and direct) · Unreal Engine 5 (C++, AI-assisted) · Hugging Face · Modal (serverless GPU) · Git / GitHub · pandas / scikit-learn · SQL (certificate level)

EXPERIENCE

Founder, Continuum Goods · Remote · 2024–Present

- Designed and directed a Python multi-agent automation stack (AI-assisted implementation) integrating Shopify, Instagram, Pinterest and Facebook APIs with LLM-driven content and decision agents.

Produce Supervisor · Costco Wholesale · Pewaukee, WI · Oct 2024–Present

- Supervise the produce department in a high-volume warehouse: team coordination, process documentation, and independent operational decisions under pressure.

Employment Specialist / Job Coach · Pantheon · WI · 2024

- Assessed client needs and built tailored support plans; coordinated across clients, employers, and case managers.

Electrician · Pieper Electric · WI · 2023–2024 · Server / Trainer · Olive Garden · WI · 2017–2022

EDUCATION

Data Science Certificate · TripleTen · Completed 2026 · Python · SQL · Machine Learning · Statistics · EDA **B.S. Criminology, Minor in Psychology · UW–Whitewater · 2016–2020**